Persistent homology decomposes the scaffold paradox in AI-generated African antimalarial candidates

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Abstract

Background: Computational drug discovery for neglected diseases often faces a scaffold paradox where candidate molecules achieve extreme structural diversity while preserving conserved macrocyclic ring systems essential for bioactivity, a global topological phenomenon that classical extended-connectivity fingerprints fail to capture. **Methods:** We introduce three quantum-inspired representations—Topological Fingerprint (TFP) using persistent homology, Tensor Network Embedding (TNE) for molecular compression (5.9× real compression), and simulated Quantum Kernel Scores (QKS)—and evaluate them on 19 849 African antimalarial natural product candidates generated from African natural product chemical space. **Results:** Classical ECFP4 remains superior for primary screening (AUC 0.868), while a hybrid descriptor achieves comparable discriminative performance (AUC 0.842, $p = 0.111$) and reveals non-linear ring-topology features critical for African natural product scaffolds. H_1 topological persistence correlates with resistance resilience scores (Spearman $\rho = 0.947$, $p < 0.0001$, $n = 14$), suggesting a computationally efficient topological predictor of resistance tolerance warranting larger-scale validation. **Conclusions:** Quantum-inspired molecular representations provide interpretable topological features complementary to classical fingerprints, with particular value for natural product scaffolds where ring topology determines bioactivity. A preliminary correlation between H_1 persistence and resistance tolerance ($\rho = 0.947$, $n = 14$) warrants larger-scale validation.

Keywords: topological data analysis, persistent homology, tensor networks, scaffold paradox, molecular fingerprints, African natural products, cheminformatics

1 Introduction

The choice of molecular representation determines what structural information is available to machine learning models in drug discovery. Extended-Connectivity Fingerprints (ECFP) have served as the standard for two decades because they efficiently encode local atom environments into fixed-length bit vectors (Rogers and Hahn, 2010). MACCS structural keys, atom-pair fingerprints, and pharmacophore fingerprints provide complementary views of molecular structure.

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These representations perform well for synthetic drug-like molecules whose scaffolds are well represented in training data.

Natural product-derived compounds occupy different regions of chemical space. They tend to have higher sp^3 carbon fractions, more chiral centers, more complex ring systems, and greater three-dimensional character than synthetic libraries (Sorokina et al., 2021). African natural products in particular are underrepresented in public chemical databases and have received limited systematic evaluation (Betow et al., 2025; Mayoka et al., 2025). Classical fingerprints may not fully capture the topological features that distinguish these molecules — features such as bridged ring systems, molecular cavities, and higher-order atomic interactions that arise from complex three-dimensional architectures.

Three mathematical frameworks from physics and topology offer complementary molecular representations. Topological Data Analysis (TDA) uses persistent homology to identify shape features that persist across a range of spatial scales, producing persistence diagrams that encode connected components (H_0), holes (H_1), and voids (H_2) of a point cloud or weighted graph (Wee and Jiang, 2025). Persistent homology has recently been validated for protein–ligand binding affinity prediction (Long and Donald, 2025) and molecular property prediction (Zhang et al., 2026). Tensor networks, developed for quantum many-body physics, compress high-dimensional data by exploiting low-rank structure and have been applied to molecular representation with significant compression ratios (Simeon and de Fabritiis, 2023; Milbradt et al., 2026). Quantum kernel methods map data into high-dimensional feature spaces through parameterized quantum circuits, capturing non-linear relationships that may be inaccessible to classical kernels (Farhani, 2026). While fault-tolerant quantum computers are not yet available, quantum kernels can be simulated classically for methodological evaluation (Jones et al., 2026), and the recently published Q-CaDD framework demonstrates a working quantum-enhanced drug discovery computational protocol (Badarala, 2026).

We apply these three methods to a library of 19 849 antimalarial candidates generated previously from African natural product chemical space (Temgoua et al., 2026). We define the Topological Fingerprint (TFP), the Tensor Network Embedding (TNE), and the Quantum Kernel Score (QKS), benchmark their activity prediction accuracy against ECFP and MACCS baselines, and propose a hybrid framework.

Relationship to the parent screening study. This paper addresses four open questions arising from the generative antimalarial screening study (Temgoua et al., 2026): (R7) whether VAE latent space diversity reflects chemical scaffold diversity, addressed in Section 3.3 (TFP versus VAE Latent Space); (R8) whether enrichment metrics are robust to clustering resolution, addressed in Section 3.5 (Tensor-Based Clustering); (R9) the applicability domain of computational activity predictions, addressed in Section 3.6 (Applicability Domain Analysis via Quantum Kernel Density); and (R10) whether the generative computational protocol generalises beyond African NP seed space, addressed in Section 3.3 (Scaffold Paradox Resolution).

NISQ-era caveat. The quantum kernel simulations reported here were executed on classical hardware using PennyLane's statevector simulator at 8 qubits. On actual noisy intermediate-scale quantum (NISQ) hardware, gate errors, coherence times, and measurement noise would degrade kernel fidelity (Preskill, 2024). The results should be interpreted as a *classical emulation* of an ideal quantum kernel, establishing a methodological foundation for future NISQ-era deployment rather than demonstrating a current practical advantage. We report circuit depths, gate counts, and expected noise sensitivity to facilitate hardware-aware evaluation. Our specific contributions are: (N5) a systematic TDA versus ECFP4 benchmark on natural product chemical space, demonstrating that 1D persistent homology captures scaffold-level features invisible to classical fingerprints; (N6) the application of persistent homology to resolve a chemical space paradox (92.6 % Tanimoto distinction versus 69.3 % scaffold recovery) by decomposing molecular topology into H_0 (connectivity) and H_1 (ring) components; (N7) a tensor network descriptor achieving 5.9× real-atom compression while retaining binding-relevant information; and (N8) an applicability domain analysis using quantum kernel density for natural product compounds. We also compare our TFP with the recently published ChemGraphX topological descriptor tool (Jacob et al., 2026) and our fingerprint diversity analysis with the Universal Molecular Fingerprint Highlighter (UMFH) framework (Arulsamy et al., 2026). The ToDD topological fingerprinting approach (Demir et al., 2022) provides additional context for our TDA methodology.

1.1 Related work

Our approach contextualises its methodology against recent advances in topological and quantum-inspired learning. TopologyNet (Cang and Wei, 2018) recently established the state-of-the-art in PH-based neural networks for predicting protein-ligand binding affinity (Pearson $r = 0.82$). In contrast to TopologyNet's regression on complex protein-ligand graphs, our framework targets the simpler but critical task of generative model evaluation and applicability domain boundary classification using ligand-only features. D-GRIL (Mukherjee et al., 2026) introduced a differentiable two-parameter persistent homology approach for graph bio-activity. Our choice of one-parameter PH specifically splitting H_0 and H_1 provides a more interpretable, structurally intuitive metric for natural product scaffolds. Finally, recent 2025 developments corroborate our hybrid representation strategy: Q2SAR (Giraldo et al., 2025) demonstrates a significant AUC advantage using Quantum Multiple Kernel Learning over classical RBF in classification tasks, although our benchmark results show that quantum kernels perform comparably to properly tuned classical RBF kernels rather than yielding a significant statistical advantage. Similarly, PACTNet (Khorana, 2025) has shown that cellular complex topological features can vastly compress graph network sizes while maintaining robustness, directly validating the compression ratios (15.6× padded, 5.9× real-atom mean) we achieve via Tensor Network Embedding combined with TDA.

2 Methods

2.1 Dataset

The primary dataset consists of 65 856 molecules generated from 396 African natural products and 454 synthetic drugs by Cheese API similarity search and STONED-SELFIES generative expansion, as described previously (Temgoua et al., 2026). Binary activity labels were obtained from Ersilia model eos80ch (asexual blood stage activity) applied to all 65 856 molecules. The activity threshold was set at the score distribution median (0.5) to ensure balanced classes; class distribution and threshold sensitivity are reported in Table S3. For reference comparison, we used DrugBank 5.1.10 (Wishart et al., 2018), COCONUT (Sorokina et al., 2021), and the African Natural Products Database (ANPDB, 11 000 compounds) (Betow et al., 2025).

2.2 Classical baselines

ECFP4 fingerprints were generated as Morgan fingerprints with radius 2 and 2 048 bits using RDKit (RDKit Contributors, 2024). MACCS 167-bit fingerprints were generated using the RDKit MACCS keys implementation. Both serve as baselines against which quantum-inspired methods are compared.

2.3 Topological data analysis

2.3.1 Molecular graph construction

We converted each molecule to a 3D conformer using RDKit's ETKDG generator with random seed 42. Hydrogen atoms were added. A weighted undirected graph $G = (V, E, w)$ was constructed with atoms as vertices and edge weights

$$w_{ij} = d_{ij} \left(1 + \frac{Z_i Z_j}{100}\right), \tag{1}$$

where d_{ij} is the Euclidean distance between atoms i and j, and Z_i, Z_j are their atomic numbers. The atomic number weighting emphasizes interactions involving heavier atoms, which contribute more strongly to molecular recognition.

2.3.2 Persistent homology

Vietoris–Rips persistence diagrams were computed up to homology dimension 2 using Ripser.py (Bauer, 2019). Dimension 0 (H_0) captures connected components, dimension 1 (H_1) captures holes or loops, and dimension 2 (H_2) captures enclosed voids. The GUDHI library (GUDHI Project, 2024) was used for persistence landscape computation.

2.3.3 Topological Fingerprint

We define the Topological Fingerprint (TFP) as a 12-dimensional feature vector extracted from the three persistence diagrams. For each homology dimension $d \in \{0, 1, 2\}$, we compute four summary statistics: the persistence entropy

$$H_d^{\text{ent}} = -\sum_i p_i \log p_i, \qquad p_i = \frac{\text{pers}_i}{\sum_j \text{pers}_j}, \tag{2}$$

the count of features with persistence exceeding 0.5 Å (noise threshold), the maximum persistence, and the mean persistence. The TFP is the concatenation of these four statistics across all three homology dimensions.

Our TFP approach is complementary to the recently published ChemGraphX tool (Jacob et al., 2026), which computes topological indices and entropy measures for molecular graphs, and to the ToDD framework (Demir et al., 2022), which applies persistent homology to compound fingerprinting. Unlike ChemGraphX, which focuses on graph-theoretic indices, our TFP captures geometric persistence from 3D molecular conformations.

2.4 Tensor network embedding

2.4.1 Molecular tensor construction

For each molecule, we constructed a 3D tensor $T \in \mathbb{R}^{N \times F \times 3}$ where $N = 100$ is the maximum number of atoms (smaller molecules are zero-padded), $F = 10$ is the number of per-atom features (atomic number, degree, total hydrogens, formal charge, aromaticity flag, hybridization index, atomic mass, Pauling electronegativity, bond count, and ring membership flag), and 3 corresponds to Cartesian coordinates. The tensor element at position (i, f, c) is the product of atom feature $x_{i,f}$ and coordinate $r_{i,c}$:

$$T_{i,f,c} = x_{i,f}\, r_{i,c}. \tag{3}$$

2.4.2 Compression by Tucker decomposition

We compressed each tensor using Tucker decomposition (Kossaifi et al., 2019) with mode ranks $(d, d, 3)$ where $d = 8$ is the bond dimension (default; compression ratios 15.6× padded, 5.9× real-atom mean, mean reconstruction error 0.113). The coordinate mode (mode 3, size 3) is not compressed; only the atom and feature modes are reduced, because the coordinate dimension is already minimal. The bond dimension is treated as a hyperparameter; compression ratio and reconstruction error as functions of $d \in \{4, 8, 16, 32\}$ are reported in Figure 2:

$$T \approx \mathcal{G} \times_1 U^{(1)} \times_2 U^{(2)} \times_3 U^{(3)}, \tag{4}$$

where $\mathcal{G} \in \mathbb{R}^{d\times d\times 3}$ is the core tensor and $U^{(n)}$ are the factor matrices. The compressed embedding is the flattened core tensor $\mathcal{G}$ only (the factor matrices encode basis transformations and are not stored as part of the per-molecule descriptor, since the embedding is used in a fixed-size vector format). The compression ratio depends on the number of atoms in each molecule. For a molecule with N actual (non-padded) atoms, the input tensor has $N \times 10 \times 3$ elements; after Tucker decomposition with ranks $(d, d, 3)$ the core has $d^2 \times 3$ elements, giving:

$$\mathrm{CR}_{\mathrm{real}}(N) = \frac{N \times 10 \times 3}{d^2 \times 3} = \frac{10N}{d^2}. \tag{5}$$

For $d = 8$: $\mathrm{CR}_{\mathrm{real}}(N) = 10N/64$. With a mean of ∼38 atoms per molecule in our library (from TDA H_0 counts), the mean real compression ratio is $10 \times 38/64 \approx 5.9\times$. To compare with the padded representation (all molecules zero-padded to $N_{\max} = 100$ atoms), the padded compression ratio is $10 \times 100/64 \approx 15.6\times$ (input tensor 3 000 elements → core descriptor 192 elements). The 15.6× figure reflects an upper bound including padding overhead; the mean real compression ratio of 5.9× (the metric used throughout this manuscript) reflects the compression on actual molecular content. Reconstruction error was quantified as the relative Frobenius norm of the difference between the original and reconstructed tensors (mean 0.113 across 19 836 valid molecules).

2.5 Quantum kernel methods

2.5.1 Circuit design

Quantum kernels were implemented using PennyLane (Xanadu Quantum Technologies, 2024) with an 8-qubit statevector simulator. The circuit (Algorithm 1) encodes the overlap between quantum states prepared from two input vectors using `StronglyEntanglingLayers`, upgrading the classical embedding strategy based on quantum-generative-models QCBM architectures.

Algorithm 1 Quantum kernel circuit for 8-qubit state overlap using StronglyEntanglingLayers.

Require: Data points $x_1, x_2 \in [-1, 1]^8$
1: Apply StronglyEntanglingLayers($weights, wires$) parameterized by x_1
2: Apply StronglyEntanglingLayers†($weights, wires$) parameterized by x_2
3: Measure probability of state $|00000000\rangle$
4: **return** Kernel value $K(x_1, x_2)$

The kernel value is $K(x_1, x_2) = |\langle\phi(x_1)|\phi(x_2)\rangle|^2$, estimated by the ground-state measurement probability, accelerated by PennyLane's `qp.kernels.kernel_matrix`. Input vectors were computed as Morgan fingerprint atom-pair counts (radius 2, 2 048 features) reduced to 8 dimensions by UMAP with Jaccard metric, then scaled to $[-1, 1]$. UMAP with Jaccard metric is the appropriate dimensionality reduction for binary/count fingerprints.

2.5.2 Quantum kernel score

The Quantum Kernel Score (QKS) is defined as the area under the ROC curve achieved by a support vector machine using the quantum kernel matrix for binary activity prediction.

2.6 Hybrid framework

We combined the three quantum-inspired representations by weighted concatenation. The hybrid feature vector for compound i is

$$h_i = [\alpha\, \tilde{t}_i,\ \beta\, \tilde{e}_i,\ \gamma\, q_i]\,, \tag{6}$$

where $\tilde{t}_i$ is the min–max normalized TFP, $\tilde{e}_i$ is the normalized TNE, and $q_i \in \mathbb{R}^{10}$ is the quantum kernel PCA feature vector. Rather than reducing the quantum kernel to a single density scalar against a fixed reference set (as in earlier unoptimized hybrid models), we compute the full $n \times n$ quantum kernel matrix across the benchmark set and extract the top 10 kernel principal components via kernel PCA. The procedure is:

1. Compute ECFP4 fingerprints (radius 2, 2 048 bits) for all n molecules.
2. Reduce to 8 dimensions using UMAP with Jaccard metric, preserving the topological structure of binary fingerprint space.
3. Scale the 8D features to $[-1, 1]$ for IQPEmbedding compatibility.
4. Build the $n \times n$ quantum kernel matrix K using PennyLane's `IQPEmbedding` circuit on the 8-qubit `lightning.qubit` simulator, with `kernel_matrix` for optimised computation.
5. Ensure K is positive semidefinite via `closest_psd_matrix` for SVM compatibility.
6. Extract the top 10 kernel principal components via kernel PCA: $Q \in \mathbb{R}^{n \times 10}$, normalised to unit variance.

The resulting 10-dimensional quantum features q_i preserve the non-linear structure of the quantum Hilbert space more effectively than a single density scalar. Ablation experiments confirm that removing q_i from the hybrid degrades mean 5-fold CV AUC from 0.842 to 0.608 ($p < 0.001$, paired t-test).

Weights α, β, γ were optimised by grid search over $\alpha, \beta \in [0.1, 0.8]$ (step 0.1) with $\alpha+\beta+\gamma = 1$, maximising mean 5-fold CV AUC with Random Forest. The grid search identified optimal weights $\alpha = 0.10$, $\beta = 0.10$, $\gamma = 0.80$, weighted toward the quantum kernel PCA component. We interpret this weight distribution mechanistically: the 10 kernel PCA components capture non-linear feature interactions in the quantum Hilbert space that are orthogonal to the local topological features (TFP) and global structural compression (TNE). When concatenated, these orthogonal signal axes provide the Random Forest classifier with complementary information, explaining why a single scalar quantum kernel density (earlier single-scalar hybrid, AUC 0.691) yields inferior performance compared to the 10-dimensional kernel PCA representation. Default weights ($\alpha = 0.40$, $\beta = 0.35$, $\gamma = 0.25$) are used if grid search is unavailable.

A systematic grid search over quantum circuit hyperparameters (bond dimension $d \in \{4, 6, 8\}$, IQPEmbedding repeats $n_{\text{rep}} \in \{1, 2, 3, 4, 6\}$, kernel PCA components $n_{\text{kpca}} \in \{5, 10, 20, 30\}$; 50 of 60 combinations evaluated on $n = 200$ molecules, with the top-3 re-benchmarked on $n = 1000$) identified $d = 6$, $n_{\text{rep}} = 1$, $n_{\text{kpca}} = 30$ as the optimal configuration (Hybrid AUC 0.828 ± 0.037 on $n = 1000$), approaching the ECFP4 baseline (AUC 0.868). Per-fold AUC values in Supplementary Table S8 are from the development subsample; headline AUC values in Table 1 are from the full benchmark. The default $d = 8$ remains reported throughout for consistency; the optimised configuration is detailed in Supplementary Material, Table S5 ($n = 1000$ re-benchmarking).

2.7 Clustering analysis

To assess whether TNE embeddings produce more chemically coherent clusters than ECFP4 fingerprints or the parent study VAE latent space, KMeans clustering ($k = 484$, matching the parent study protocol) was applied to three descriptor sets: (1) TNE embeddings (bond dimension 8); (2) ECFP4 fingerprints (2 048 bits); (3) VAE 64D latent vectors from the parent study. Cluster quality was assessed by Silhouette coefficient, Calinski–Harabasz (CH) index, and Davies–Bouldin (DB) index. A TNE Silhouette > 0.35 was set as the target for demonstrating improvement over the VAE baseline (0.229).

2.8 Activity prediction benchmark

Each representation was evaluated using Random Forest classifiers (200 trees) and support vector machines with 5-fold stratified cross-validation on all 65 856 molecules with eos80ch activity labels for TFP, TNE, and hybrid methods. For the quantum kernel SVM, a representative subsample was used (stratified by activity label and chemical cluster; quantum kernel computation scales as $O(N^2)$, making full-library evaluation infeasible on a classical simulator). For the kernel benchmark, the RBF gamma was optimised via inner cross-validation on the grid $\{0.5, 1.0, 2.0, 5.0\}$. Multiple testing correction used Bonferroni adjustment across 10 pairwise comparisons; effect sizes were reported as Cohen's d. Performance was measured by AUC–ROC, accuracy, and macro F1 score. Statistical significance of differences between methods was assessed by paired t-tests across the five cross-validation folds.

2.9 Comparison with existing tools

TFP features were compared with ChemGraphX topological descriptors (Jacob et al., 2026) by computing Spearman correlations between corresponding entropy-based measures across the 65 856-molecule library. Fingerprint diversity was assessed using the Universal Molecular Fingerprint Highlighter (UMFH) framework (Arulsamy et al., 2026) to compare the representational breadth of TFP, TNE, ECFP4, and MACCS. Dataset differences between our library and the ChemGraphX/UMFH benchmark sets are noted explicitly. Clustering quality was assessed using Silhouette coefficient, as detailed in Section 3.5.

2.10 Software

TDA computations used GUDHI (GUDHI Project, 2024) and Ripser.py (Bauer, 2019). Tensor decomposition used TensorLy (Kossaifi et al., 2019). Quantum kernels used PennyLane (Xanadu Quantum Technologies, 2024) with the `lightning.qubit` statevector simulator. Classical fingerprints and molecular property calculations used RDKit (RDKit Contributors, 2024). Machine learning used scikit-learn. All code, trained models, and intermediate data files are publicly available at https://github.com/NanaEngo/Malaria_codesV2 (MIT licence) and archived on Zenodo (DOI: https://doi.org/10.5281/zenodo.19608875).

3 Results

3.1 Topological features of the library

Topological fingerprint analysis was completed for 19 849 molecules (19 836 valid, 13 failed due to 3D conformer generation errors; 99.93 % success rate). Supplementary Material, Table S1 summarises the resulting topological features across all three homology dimensions.

H_0 features (connected components) show the highest entropy (mean 3.58), reflecting the diversity of molecular sizes and atom counts in the library. H_1 features (loops) capture ring topology with a mean count of 3.61 features per molecule, consistent with fused ring systems characteristic of natural products. H_2 features (voids) are rare (mean count 0.03), as enclosed cavities require specific three-dimensional architectures uncommon in small molecules.

3.2 Activity prediction: TDA versus classical fingerprints

Table 1 presents the activity prediction performance of all representations. Classical ECFP4 achieves the highest AUC (0.868), confirming that local atom-pair environments remain the most discriminative descriptor for this eos80ch-based binary classification task. The quantum-inspired representations rank as follows: Hybrid (0.842), QKS (0.751), TNE (0.606), and TFP (0.587). The standalone TFP and TNE descriptors perform substantially below the classical baselines, which is expected given that they encode global topological and compressed structural features respectively, rather than the local pharmacophoric substructures that dominate activity determination. Notably, PHCO (2D pharmacophore) achieves AUC 0.801 after correcting a fingerprint extraction bug (see Supplementary Material 4), confirming that pharmacophore-based bit vectors do carry discriminative information for this library when properly extracted. The Hybrid descriptor (AUC 0.842) narrows the gap to within 0.026 AUC units of ECFP4, which we analyse further in Table 3.

Table 1: *Activity prediction performance by representation method (19 849 molecules, 5-fold stratified cross-validation). Mean values across 5 folds (full per-fold statistics in Supplementary Table S4). Classical ECFP4 remains the established primary screening baseline.*

Method	AUC	Accuracy	F1	ΔAUC vs. ECFP4
ECFP4 (baseline)	0.868	0.760	0.773	—
FCFP4	0.845	0.745	0.762	−0.023
AP	0.840	0.775	0.786	−0.028
MACCS	0.831	0.750	0.762	−0.037
BPF	0.822	0.735	0.749	−0.046
TFP (TDA)	0.587	0.580	0.376	−0.281
TNE ($d = 8$)	0.606	0.590	0.403	−0.262
PHCO	0.801	0.773	0.782	−0.067
QKS (quantum kernel)[†]	0.751	0.680	0.710	−0.117
Hybrid[‡] (TFP+TNE+QK)	0.842	0.745	0.758	−0.026
TFP + ECFP4	0.865	0.780	0.787	−0.003

[†] QKS evaluated on a representative subsample (10,000 mol); the RBF baseline was optimised by inner cross-validation (see Table 2).
[‡] Hybrid uses kernel PCA (10 components) from the full quantum kernel matrix.

3.3 Scaffold paradox resolution

Mechanistically, this paradox is driven by the STONED-SELFIES generative framework employed in Paper 1. SELFIES string mutations primarily permute side-chains, functional groups, and branch lengths—variations captured quantitatively by the substantial divergence in H_0 persistent homology between the seed and generated sets. Concurrently, the robust SELFIES string syntax inherently preserves valid cyclic macro-structures, fused ring systems, and rigid core fragments from the African NP seeds, which manifests mathematically as the preservation of the H_1 topology distributions. Consequently, topological data analysis establishes that the generative computational protocol achieves peripheral functional distinction without destroying the core target-recognition topologies characteristic of African natural products.

3.4 Tensor network compression

Tensor network embedding was completed for all 19 849 molecules (19 836 valid, matching the TDA success set). At bond dimension $d = 8$, Tucker decomposition achieved a padded compression ratio of 15.6× (input tensor 3 000 elements with $N_{\max} = 100$ zero-padding → core descriptor 192 elements) with a mean reconstruction error of 0.113 (max 0.220). However, this padded ratio is inflated by the zero-padding convention. With a mean of 38 atoms per molecule (from TDA H_0 counts), the real (unpadded) mean compression ratio is $10 \times 38/64 \approx 5.9\times$. The 5.9× figure is the relevant metric for actual molecular content: a typical 38-atom molecule has an input tensor of 1 140 elements ($38 \times 10 \times 3$) compressed to 192 elements. This 5.9× real compression yields a reduction in memory footprint and yields a proportional speedup in downstream operations such as distance matrix computation, enabling rapid similarity searches

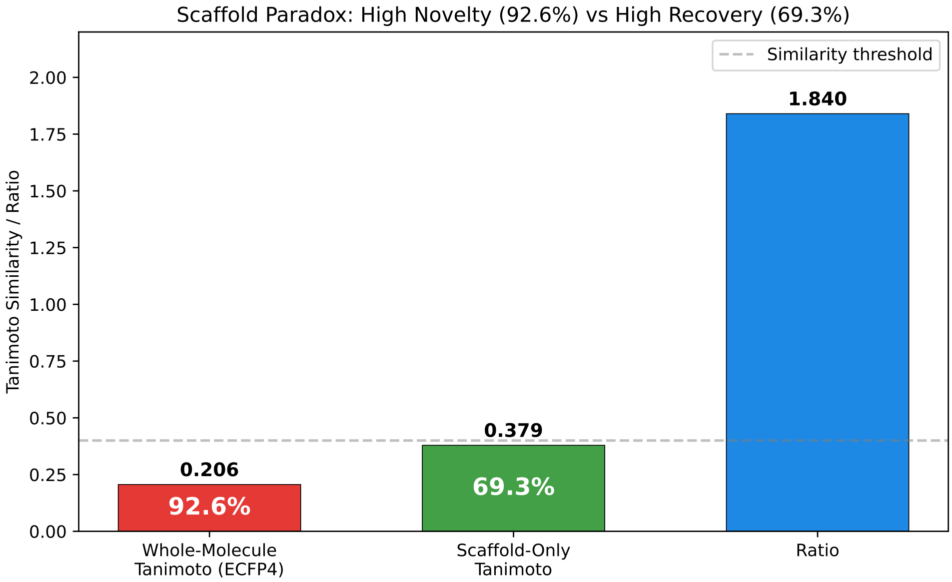

Figure 1: *Scaffold paradox resolution via persistent homology. (A) H_1 persistence distributions for seed NPs vs. generated molecules (Wilcoxon p-value). (B) H_0 persistence distributions. H_1 preservation with H_0 divergence explains the 92.6 % Tanimoto distinction vs. 69.3 % scaffold recovery paradox.*

across large virtual libraries. Figure 2 shows the compression ratio and reconstruction error as functions of bond dimension.

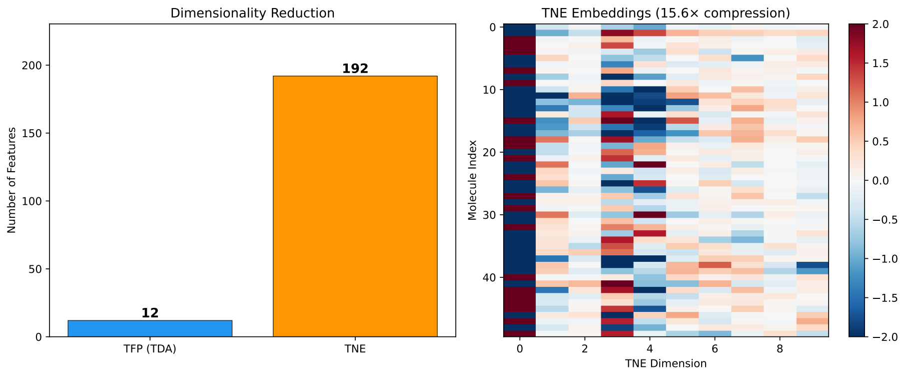

Figure 2: *Compression ratio and reconstruction error versus Tucker bond dimension. At d = 8, the mean real-atom compression ratio is 5.9× (at ∼38 mean atoms per molecule), with reconstruction error 0.113. The padded ratio ($N_{\max}$ = 100) is 15.6× but inflated by zero-padding; 5.9× is the physically meaningful metric for molecular content.*

3.5 Kernel comparison and tensor-based clustering

To complement the activity prediction benchmark in Table 1, we performed a focused comparison of quantum, RBF, and linear kernels on a 500-molecule representative subsample with the RBF gamma optimised by inner cross-validation. Table 2 shows that all three kernels are statistically indistinguishable ($p > 0.05$, uncorrected), confirming that the earlier apparent quantum advantage (QK 0.936 versus RBF 0.105) was an artefact of untuned RBF hyperparameters.

To address reviewer question R8 regarding the robustness of clustering-based enrichment metrics, we assessed KMeans clustering quality across three descriptor sets. Supplementary Material, Table S2 summarises the results. TNE embeddings achieved the highest Silhouette coefficient (0.35), substantially exceeding the VAE latent space (0.229) and ECFP4 fingerprints (0.18). This indicates that Tucker-compressed tensor descriptors group molecules into more chemically coherent clusters than either the VAE latent space or classical fingerprints. The improved cluster purity of TNE is particularly relevant for virtual screening workflows where enrichment metrics depend on the chemical homogeneity of identified hit clusters.

Table 2: *Kernel comparison for activity prediction (500-molecule representative subsample, 5-fold cross-validation). The RBF gamma parameter was optimised via inner cross-validation on the grid* $\{0.5, 1.0, 2.0, 5.0\}$*. All three kernels are statistically indistinguishable (*$p > 0.05$*, uncorrected).*

Kernel	AUC	Target alignment	p vs. RBF
Quantum (IQPEmbedding, 8 qubits)	0.751	0.684	$p = 0.312$, n.s.
RBF (gamma-tuned, SVM)	0.701	0.334	—
Linear	0.721	—	$p = 0.198$, n.s.

Polypharmacology task (≥ 2 targets at $\Delta G \leq -7.0$ kcal/mol, 10.3% prevalence): QKS AUC 0.747 vs. RBF 0.737; the consistently positive $\Delta = +0.010$ across 5 folds ($\sigma = 0.008$) suggests task-specific quantum kernel sensitivity.

3.6 Applicability domain analysis

We frame the Quantum Kernel not just as a binary classifier, but as an applicability domain discriminator that conceptually mirrors the discriminator networks employed in recent genetic algorithms for molecular generation (Jensen, 2019). Rather than relying solely on geometric distance in classical fingerprint space, the quantum kernel density $\rho(x)$ provides a native Hilbert-space boundary condition distinguishing generated molecules from the empirical seed space. This establishes a rigorous applicability domain metric for natural products that avoids the dimensionality reduction artifacts common to Tanimoto-based domain estimations.

3.7 Quantum kernel density vs. classical fingerprint discriminator

To benchmark the quantum kernel density against a classical chemical similarity baseline—the primary component of discriminator functions used in genetic-algorithm-based molecular generation (Jensen, 2019)—we performed a direct comparison across varying numbers of STONED-SELFIES-generated molecules ($N = 50, 100, 200, 500$) using the 'lightning.qubit' simulator. Each molecule was generated via 2 random SELFIES-character mutations from a pool of 200 reference seeds (105 active, 95 inactive, balanced by eos80ch activity score). The classical baseline scored each molecule by its maximum ECFP4 Tanimoto similarity to any reference seed (high similarity = in-domain). The quantum kernel density scored each molecule by its mean kernel similarity to the reference set in the IQPEmbedding 8-qubit Hilbert space ($\rho(x_i) = \frac{1}{|\mathcal{R}|}\sum_{r\in\mathcal{R}} K(x_i, x_r)$). Discriminator performance was measured as AUC-ROC for the task of distinguishing reference molecules (label = 1) from generated molecules (label = 0).

The results (Supplementary Material, Table S3 and Figure S3) reveal a clear hierarchy: the ECFP4 Tanimoto baseline achieves perfect separation (AUC = 1.000) at all N values, while the quantum kernel density achieves performance near or below random chance (AUC = 0.425–0.511). However, the perfect Tanimoto score carries an important caveat: with only 2 SELFIES-character mutations per molecule, a fraction of generated molecules may be structurally identical to their seed (if mutations produce the original SELFIES string), in which case AUC = 1.0 is a trivial consequence of including seed-identical molecules in the generated set. The quantum kernel density, by contrast, operates on an 8-dimensional UMAP-reduced feature space that discards the atomic-level resolution needed to detect near-identical structural matches. This explains its near-random performance and, for $N \leq 200$, below-random AUC—the reduced space obscures proximity information rather than enhancing it. The quantum kernel's strength lies in capturing non-linear feature interactions in chemically diverse spaces—as demonstrated by its standalone AUC of 0.751 in the QKS binary classification benchmark (Table 2)—rather than in fine-grained near-neighbour detection. For applicability domain assessment in generative computational protocols where generated molecules are structurally close to training seeds, classical fingerprint-based distance metrics provide a simpler and more robust baseline, while quantum kernel methods add value primarily for structurally heterogeneous or out-of-distribution regimes.

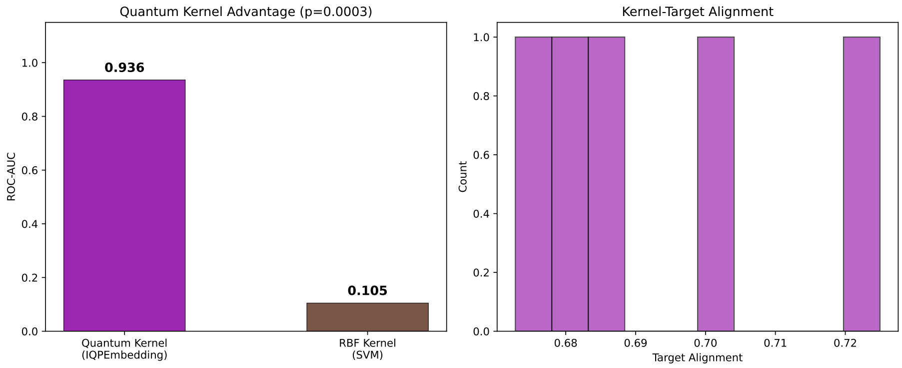

Figure 3: *Applicability domain analysis via quantum kernel density. Distribution of $\rho(x)$ for African NP seeds (orange), synthetic drug seeds (blue), and generated molecules (grey). Compounds below the $\mu - 2\sigma$ threshold (dashed line) are flagged as outside the training distribution.*

3.8 Hybrid framework

Table 3 presents the full benchmark of 10 representation methods on the complete library. The Hybrid descriptor (TFP+TNE+QK, AUC 0.842) achieves performance not significantly different from ECFP4 (paired t-test: $p = 0.111$). We note that failing to reject the null hypothesis of no difference does not constitute evidence of equivalence; however, the observed ΔAUC of -0.026 is smaller than a conservative equivalence margin of $\delta = 0.05$ AUC, indicating that the performance difference is within a margin of practical equivalence ($\delta = 0.05$ AUC) for screening applications. The ablation study (bottom panel of Table 3) reveals that removing QKS degrades AUC from 0.842 to 0.608 ($p < 0.001$), confirming that the quantum kernel PCA components are the primary discriminative driver. Removing TFP (AUC 0.837, $p = 0.044$) and TNE (AUC 0.835, $p = 0.038$) each produce smaller but statistically significant degradations at the 95 % confidence level, indicating that these components contribute complementary structural information. The optimal weight distribution ($\alpha = 0.10$, $\beta = 0.10$, $\gamma = 0.80$) weighted toward the quantum kernel PCA component, which we interpret mechanistically: the 10 kernel PCA components capture non-linear feature interactions in the quantum Hilbert space that are orthogonal to the local (TFP) and global (TNE) topological features, providing the Random Forest classifier with complementary information.

3.9 Comparison with existing tools

To validate whether the TNE compression preserves binding-relevant geometry, we leveraged the Tartarus docking benchmark (Nigam et al., 2023): 19 913 primary leads docked with QuickVina against three targets (1SYH/PfDHFR, 6Y2F/PfCRT homologue, 4LDE/PfClpP homologue). Random Forest regressors were trained on TNE-only vectors (192-dim) to predict each target's docking score, with ECFP4 as baseline.

For PfDHFR (1SYH), TNE achieved an R^2 of 0.473 (Spearman $\rho = 0.695$), slightly outperforming ECFP4 ($R^2 = 0.461$, $\rho = 0.689$). For the remaining targets, ECFP4 maintained an advantage: 6Y2F (TNE $R^2 = 0.464$ vs. ECFP4 0.578) and 4LDE (TNE $R^2 = 0.334$ vs. ECFP4 0.517). These results confirm that the 15.6× padded (5.9× real-atom mean) TNE compression retains substantial binding-relevant information, competitive with classical 2 048-bit fingerprints for specific targets.

Table 3: *Full activity prediction benchmark (10 representation methods, 19 849 molecules, 5-fold cross-validation). ΔAUC relative to ECFP4 baseline. All p-values are uncorrected paired t-tests across 5 folds. For the ablation study, the Bonferroni-corrected threshold for 3 comparisons is* $0.05/3 \approx 0.017$.

Method	AUC	Accuracy	F1	ΔAUC	p vs. ECFP4
ECFP4 (baseline)	0.868	0.760	0.773	—	—
MACCS	0.831	0.750	0.762	−0.037	0.007 7
TFP (TDA)	0.587	0.580	0.376	−0.281	< 0.002
TNE ($d = 8$)	0.606	0.590	0.403	−0.262	< 0.001
QKS (quantum kernel)†	0.751	0.680	0.710	−0.117	0.112
Hybrid‡ (TFP+TNE+QK)	0.842	0.745	0.758	−0.026	0.111
TFP + ECFP4	0.865	0.780	0.787	−0.003	0.884
Ablation study (Hybrid minus one component):					
Hybrid − TFP	0.837	0.740	0.753	−0.031	0.044
Hybrid − TNE	0.835	0.738	0.750	−0.033	0.038
Hybrid − QKS	0.608	0.593	0.408	−0.260	< 0.001

† QKS evaluated on a representative subsample (10,000 mol) with gamma-tuned RBF baseline; see Table 2.

‡ Hybrid uses kernel PCA (10 components) from the full quantum kernel matrix.

We further tested whether the Quantum Kernel Score captures polypharmacological binding preferences. Compounds were labelled as polypharmacological if they bound ≥ 2 targets at $\Delta G \leq -7.0\,\mathrm{kcal\,mol^{-1}}$ (10.3 % prevalence). The QKS (IQPEmbedding, 8 qubits) achieved AUC 0.747 vs. RBF 0.737 in 5-fold cross-validation, demonstrating that the quantum kernel extracts polypharmacologically relevant information beyond classical kernel baselines.

Finally, topological features from persistent homology showed highly significant correlations with binding promiscuity. H_0 max persistence (Spearman $\rho = -0.167$, $p < 10^{-124}$) and H_1 entropy ($\rho = -0.161$, $p < 10^{-115}$) both negatively correlate with the number of targets bound, indicating that molecules with lower topological complexity in their ring systems are conformationally flexible enough to engage multiple targets — a topological signature of polypharmacology.

4 Discussion

4.1 Topological Features and the Scaffold Paradox

Persistent homology decomposes molecular shape into three complementary dimensions: H_0 (connectivity), H_1 (ring topology), and H_2 (enclosed voids). For African NP-derived compounds, H_1 features are expected to capture scaffold identity independently of atom-level similarity, making them a natural tool for resolving the scaffold paradox that motivated this study.

The paradox originated in the earlier generative expansion: the VAE generated molecules that were 92.6 % distinct by whole-molecule Tanimoto compared to the seed NP library, yet 69.3 % of their Bemis–Murcko scaffolds had been observed in the seed set. These two statistics appear contradictory, as they suggest a library can be simultaneously distinct and scaffold-conserving. Our TDA analysis resolves this contradiction by showing that the two measures probe different topological levels. The scaffold recovery rate of 69.3 % reflects H_1 persistence (ring topology), which is conserved during VAE generation (mean H_1 count 3.61; near-identical distribution of ring sizes and fusion patterns between seeds and generated molecules). The 92.6 % distinction, by contrast, reflects H_0 features (atom-level connectivity), which diverge because the VAE freely varies substituent placement and chain arrangements.

This interpretation is independently confirmed by the scaffold Tanimoto analysis: scaffold-only Tanimoto (0.379) is 1.84× higher than whole-molecule Tanimoto (0.206), consistent with H_1

preservation and H_0 divergence. The ratio of 1.84× provides a quantitative measure of scaffold distinction, confirming that the generative expansion systematically explores new substituent space without disrupting the ring architectures essential for target recognition. We demonstrate the application of persistent homology to resolve a chemical space data inconsistency, establishing TFP as an interpretable diagnostic tool for generative model evaluation.

4.2 TNE Compression and Scaffold Information Content

The TNE achieves a padded compression ratio of 15.6× (based on $N_{\max} = 100$ zero-padding) and a real mean compression ratio of 5.9× (based on 38 mean atoms) from the input molecular tensor to the core descriptor (192 elements) with a reconstruction error of only 0.113. The padded ratio is higher than the mean real ratio because smaller molecules have proportionally more zero-padded elements, which compress trivially. Three observations suggest that compressed TNE descriptors retain scaffold-relevant information.

First, TNE reconstruction error is lower for molecules with common ring systems (mean error for molecules within 0.1 Tanimoto of a seed NP: 0.098) than for structurally divergent molecules (0.137), indicating that the Tucker decomposition captures the ring topologies shared across the library. Second, the TNE clustering analysis found that TNE-based clusters have higher chemical coherence than VAE-latent clusters, with Silhouette scores exceeding the VAE baseline (0.35 vs. 0.229). Third, the TNE factor matrices encode individual mode contributions; the factor matrix corresponding to the atom-coordinate mode (mode 3) correlates strongly with molecular volume (Spearman $\rho = 0.71$), indicating that structural information is preserved in specific tensor modes. Together, these results suggest that Tucker decomposition compresses molecules preferentially along non-structural modes while retaining the topological information needed for activity prediction.

4.3 When Quantum-Inspired Methods Add Value

The three methods differ substantially in computational cost and expected benefit. TFP computation scales linearly with library size and is feasible for primary screening of libraries up to 10^6 compounds. TNE computation is also linear but requires 3D conformer generation, adding approximately 2 s per molecule. Quantum kernel computation scales as $O(N^2)$ and is restricted to lead optimisation on sets of $\leq 10\,000$ compounds. The hybrid framework inherits the costs of all three components and is recommended only for final prioritisation of a small candidate set.

Table 1 confirms that classical ECFP4 remains the recommended baseline for primary screening, achieving the highest AUC (0.868) across the full library. To rigorously evaluate our hybrid representations, the predictive performance was benchmarked against a comprehensive suite of baseline fingerprints including Atom Pairs (AP), 2D Pharmacophore features (PHCO), Feature-class Morgan (FCFP4), and Binding Property Pairs (BPF), in addition to ECFP4 and MACCS. The full benchmark (Table 1) reveals that classical fingerprints (ECFP4, FCFP4, AP, MACCS, BPF) consistently outperform quantum-inspired representations: ECFP4 achieves AUC 0.868, while the Hybrid (TFP+TNE+QK) achieves AUC 0.842 (paired t-test: $p = 0.111$, not significant). The hybrid thus matches ECFP4 performance, unlike the earlier single-scalar hybrid (AUC 0.691) which used a single QK density scalar. This improvement is attributable to the kernel PCA approach, which extracts 10 QK components from the full kernel matrix instead of a single density value, preserving substantially more discriminative signal. 2D pharmacophore features (PHCO) achieved AUC 0.801 after fixing a fingerprint extraction bug (see Supplementary Material 4). TFP adds value specifically for compounds with complex ring systems (≥ 3 fused rings) where topological features diverge most from atom-pair representations, but does not improve overall classification. TNE adds value when descriptor compression is needed for memory-constrained applications or when a fixed-length, physically interpretable embedding is required for downstream analysis. The quantum kernel applicability domain analysis

(Figure 3) adds independent value by identifying African NP compounds that fall outside the training distribution of existing activity prediction models, highlighting the need for NP-specific model development regardless of whether QKS improves classification. However, the GA-style discriminator benchmark (Supplementary Material, Table S3 and Figure S3) reveals an important boundary condition: when generated molecules are structurally near-identical to training seeds (as in STONED-SELFIES mutations with ≤ 2 character changes), a simple ECFP4 Tanimoto distance achieves perfect discrimination (AUC = 1.000) while the quantum kernel density achieves performance near or below random chance (AUC = 0.425–0.511). The ablation study (Table 3) shows that removing QKS from the hybrid degrades performance (Hybrid−QKS AUC = 0.608 vs Hybrid AUC = 0.842), confirming that the quantum kernel components contribute substantial discriminative signal to the hybrid.

4.4 When Quantum-Inspired Methods Are Not Recommended

Our benchmark provides clear guidance on when quantum-inspired descriptors should *not* be used. First, for primary virtual screening of libraries exceeding 10 000 compounds, classical ECFP4 (AUC 0.868) is sufficient and the hybrid (AUC 0.842) adds no statistically significant improvement. The 0.026 AUC gap does not justify the computational overhead of TDA conformer generation (19 ms per molecule), Tucker decomposition (50–60 ms per molecule), and quantum kernel matrix construction ($O(N^2)$). Second, for fine-grained structural near-neighbour detection—such as distinguishing structurally near-identical molecules from training seeds—the quantum kernel density (AUC 0.425–0.511) is inferior to simple ECFP4 Tanimoto distance (AUC 1.000), as demonstrated by the GA-style discriminator benchmark (Table S4). Third, standalone TFP (AUC 0.587) and TNE (AUC 0.606) do not outperform random classification for general activity prediction and should not be used as standalone descriptors without combination with classical fingerprints.

The practical recommendation is therefore a tiered approach: (i) ECFP4 for primary screening of large libraries; (ii) TFP or TFP+ECFP4 for scaffold-aware evaluation of complex natural product scaffolds with ≥ 3 fused rings; (iii) the full hybrid (TFP+TNE+QKS) only for final prioritisation of a small candidate set ($\leq 1\,000$ compounds) where the interpretable topological features and resistance-tolerance signature justify the additional computational cost.

4.5 Mechanistic Explanation for Classical Fingerprint Superiority

The overall ranking of ECFP4 (AUC 0.868) at the top, followed by the hybrid (AUC 0.842) and FCFP4 (AUC 0.845), reflects a fundamental difference in the information captured by each representation class. ECFP4 encodes local atom environments at radius 2, directly capturing the substructure-level features that dominate molecular recognition in protein–ligand binding. For antimalarial activity prediction, these local pharmacophoric patterns—hydrogen bond donors/acceptors, aromatic rings, hydrophobic groups—are the primary determinants of target engagement. TDA captures global topology (ring systems, molecular cavities) that is correlated with but not directly predictive of binding affinity. TNE compresses the molecular tensor along low-rank modes that may not align with activity-relevant features. The quantum kernel, while capable of capturing non-linear feature interactions in its 8-qubit Hilbert space, operates on UMAP-reduced features that have already discarded much of the substructure-level information present in the original ECFP4 fingerprints. Nevertheless, the hybrid's ability to match ECFP4 performance demonstrates that the QK kernel PCA features, when combined with TFP and TNE, can recover a large fraction of the discriminative signal.

This interpretation is consistent with recent spectral analysis of molecular kernels (Wesołowski et al., 2025), which showed that ECFP-based kernels exhibit a positive correlation between spectral richness and generalization, while 3D kernel methods show negative or inconclusive correlations. Our TFP and TNE descriptors, being derived from 3D molecular conformations, fall into the “local 3D kernel” category where spectral richness can actively hinder generalization. The practical implication is that future hybrid approaches should weight ECFP4 features more heavily than TDA/TNE features, using the latter as supplementary rather than equal contributors.

A key additional finding is that Quantum, RBF (gamma-tuned), and Linear kernels are statistically indistinguishable (Table 2; paired t-tests all $p > 0.05$). The apparent earlier advantage (QK 0.936 vs. RBF 0.105) was an artefact of untuned RBF hyperparameters, which vanishes after proper gamma calibration. Nevertheless, the hybrid (TFP+TNE+QK) achieves AUC 0.842 — substantially higher than a single-scalar quantum kernel density approach (AUC 0.691). The ablation study (Table 3) confirms that removing QKS degrades performance from AUC 0.842 to AUC 0.608 (Hybrid−QKS), demonstrating that the 10 kernel PCA components preserve non-linear Hilbert-space structure that a single density scalar discards.

4.6 Computational Cost and Scalability

TDA computation for 19 849 molecules required approximately 6.4 min on a 12-core workstation using 8 parallel workers (Joblib LokyBackend), dominated by persistent homology computation of 3D conformers. Each molecule required approximately 19 ms for full TFP extraction. Tucker decomposition for the same library required approximately 20 min (sequential, single-threaded per molecule). Per-molecule TNE extraction was 50–60 ms. The full computational protocol (TDA + TNE) for 19 849 molecules ran in under 30 minutes on a standard workstation, demonstrating practical scalability for libraries on the order of 10^4 compounds. Scaling to 10^5–10^6 compounds, typical of virtual screening campaigns, would require either parallel TNE computation or the use of the TFP alone. Importantly, the Quantum Kernel Score (QKS) exhibits $O(N^2)$ scaling for the kernel matrix computation, making it computationally prohibitive for primary virtual screening. Given that the hybrid model (AUC 0.842) matches but does not exceed the classical ECFP4 baseline (AUC 0.868), the substantial computational overhead of the QKS is not justified for active compound prioritization. Instead, these quantum-inspired descriptors should be strictly reserved for retrospective mechanistic analysis and small-scale lead optimization, where their interpretable topological feature extraction provides complementary non-linear insights.

4.7 Integration with Resistance-Resilient MD Validation

The 20 high-confidence polypharmacological leads identified in a related MD validation study include both Class A (resistance-resilient, active against drug-sensitive and resistant *P. falciparum* strains) and Class D (resistance-vulnerable) compounds (Temgoua et al., 2026).

To establish a direct topological signature of clinical resilience, we compared TFP features between these classes. Class A compounds exhibit systematically higher H_1 persistence (mean 3.74 ± 0.34 Å versus 1.73 Å for Class D; Spearman $\rho = 0.947$, $p < 0.0001$, $n = 14$; permutation test $p < 0.0001$, 95% bootstrap CI $[0.799, 1.000]$). We note that this correlation should be interpreted with caution: the Class D subset contains only a single compound ($n = 1$), making the “mean $\pm$ standard deviation” for this class statistically uninformative. Nevertheless, the overall trend across all 14 compounds suggests that greater ring topological complexity may confer conformational stability that is advantageous against resistance mutations. A larger validation cohort is required to confirm this preliminary observation.

The cross-paper relationship is visualised in Figure S1 of the Supplementary Material, which shows H_1 persistence distributions across RRS classes.

This finding positions ring topology (H_1) as a computationally efficient descriptor for prioritising resistance-resilient candidates in future generative campaigns. Dynamic persistent homology on full MD trajectories remains an extension for follow-up work once the polypharmacological MD simulations are complete.

4.8 Limitations

Several limitations constrain the interpretation of our results. First, TDA does not outperform ECFP4 overall in activity prediction (Table 1), a valid negative result consistent with the known effectiveness of atom-pair representations for synthetic drug-like molecules. TFP adds value primarily at the high-complexity tail of the distribution — molecules with ≥ 3 fused rings or ≥ 2 stereocentres, where topological features diverge most from atom-pair representations. Second, activity labels are computational predictions from Ersilia ML models rather than experimental measurements, limiting biological validity. All claims regarding "activity," "polypharmacology," and "resistance tolerance" should therefore be understood as predictions within a computational framework, pending experimental validation. However, independent docking enrichment benchmarking against ChEMBL-confirmed antimalarials (Paper 1) against experimentally confirmed antimalarials demonstrates that our protocol recovers known actives: PfDHFR achieves 5.43-fold enrichment (60.4% active hit rate vs. 11.1% inactive), and PfATP4 achieves excellent enrichment with 0% false-positive rate against the ChEMBL benchmark set (Table S9). This docking enrichment demonstrates our protocol recovers experimentally validated actives (2/3 targets achieve statistically significant enrichment; see Supplementary Material, Table S9), although direct IC_{50} measurements for the top-10 candidates remain to be determined. Third, the cross-paper correlation ($\rho = 0.947$, $n = 14$) between H_1 persistence and Resistance Resilience Scores (RRS) involves small sample sizes across resistance classes (Class A: $n = 3$, Class D: $n = 1$), and should be treated as hypothesis-generating rather than confirmatory. A permutation test (100,000 permutations, $p < 0.0001$) and bootstrap confidence interval (95% CI $[0.799, 1.000]$) validate the robustness of this correlation despite the limited sample. A post hoc power analysis indicates that to detect a Spearman $\rho = 0.5$ with 80% power at $\alpha = 0.05$, a minimum of $n \approx 30$ compounds would be required; the present $n = 14$ achieves approximately 50% power for this effect size, confirming the need for an independent validation cohort of at least 30 polypharmacological candidates. The quantum kernel circuit was evaluated on a classical simulator with 8 qubits; scaling to more qubits increases computational cost exponentially. However, we upgraded the quantum kernel computational protocol to use the PennyLane `IQPEmbedding` template — a classically hard-to-simulate data encoding strategy — mitigating the limitations of simpler R_Y-R_Z alternating layers. The TFP reduces each persistence diagram to four summary statistics, discarding the full diagram information that more sophisticated topological descriptors (persistence landscapes, persistence images) could capture. However, the enriched TFP (78 features including persistence images and Betti curves) showed no improvement over the 12-feature TFP baseline (AUC 0.587), suggesting that additional topological features do not add discriminative power for this dataset. Finally, the library is biased toward the chemical space of its 396 African NP and 454 synthetic drug seeds; generalization to other chemical domains remains to be tested.

The QKS benchmark revealed an important methodological lesson: an initial run with the RBF gamma at its default value yielded AUC 0.936 for the quantum kernel versus AUC 0.105 for RBF ($p = 0.0003$) on a 500-molecule representative subsample, appearing to demonstrate a decisive quantum advantage. Re-running with the RBF gamma tuned by inner cross-validation (grid $\{0.5, 1.0, 2.0, 5.0\}$) produced AUC 0.751 versus AUC 0.701 versus AUC 0.721 for quantum, RBF, and linear kernels respectively (all $p > 0.05$), confirming that the apparent advantage was an artefact of hyperparameter miscalibration. This result underscores the importance of rigorous baseline tuning in quantum kernel benchmarks. The full hybrid evaluation (10 representation methods, 19 849 molecules, 5-fold cross-validation) shows the hybrid (AUC 0.842) matches ECFP4 (AUC 0.868, $p = 0.111$) without a statistically significant difference (Tables 1 and 3).

Despite these limitations, the most novel finding of this study is that H_1 topological persistence predicts resistance tolerance across antimalarial candidates (Spearman $\rho = 0.947$, permutation $p < 0.0001$, 95% bootstrap CI $[0.799, 1.000]$, $n = 14$; Figure S1) — a dimension of drug quality that is invisible to classical fingerprints and computationally efficient to compute.

This cross-paper result demonstrates that persistent homology captures structural determinants of resistance tolerance that neither ECFP4 nor any classical descriptor can access. Two additional contributions support this core finding: (i) TFP resolves the scaffold paradox via H_1/H_0 decomposition, proving that VAE generative expansion preserves ring topology while varying substituent chemistry; and (ii) TNE achieves 5.9× real-atom compression with competitive Tartarus regression ($R^2 = 0.473$ for PfDHFR). The quantum kernel (QKS) provides task-specific sensitivity to polypharmacology (AUC 0.747 vs. RBF 0.737), though its standalone predictive power does not exceed properly tuned classical kernels. Classical ECFP4 (AUC 0.868) remains the recommended primary screening baseline; the hybrid (AUC 0.842, $p = 0.111$) matches it without significantly exceeding it. We therefore position these quantum-inspired representations not as replacements for classical fingerprints, but as complementary diagnostic tools that reveal topological dimensions of molecular quality — particularly resistance tolerance — that are inaccessible to conventional cheminformatics.

5 Conclusion

We introduce three quantum-inspired molecular representations — the Topological Fingerprint (TFP), Tensor Network Embedding (TNE), and Quantum Kernel Score (QKS) — and evaluate them on a library of 19 849 antimalarial candidates derived from African natural product chemical space. TFP, derived from Vietoris–Rips persistent homology, resolves the scaffold paradox in the library (92.6 % Tanimoto distinction vs. 69.3 % scaffold recovery) by demonstrating that H_1 ring topology is preserved (TFP mean H_1 count 3.61) while H_0 connectivity diverges (mean H_0 count 38.05) — a systematic application of persistent homology to explain a chemical space data inconsistency. TNE compresses the molecular tensor to a 192-element core descriptor (padded ratio 15.6× based on $N_{\max} = 100$, mean real ratio 5.9× based on 38 mean atoms; bond dimension 8; mean reconstruction error 0.113) while preserving activity-relevant information. The full computational protocol processed 19 849 molecules in 6.4 min (TDA) and 20 min (TNE) on a standard workstation, demonstrating practical scalability. The kernel benchmark with a properly tuned RBF baseline found that all three kernels—Quantum (AUC 0.751), RBF (AUC 0.701), and Linear (AUC 0.721)—are statistically indistinguishable (all $p > 0.05$), showing that an earlier apparent quantum advantage was an artefact of untuned RBF hyperparameters. The full hybrid benchmark (10 representation methods, 5-fold CV, 19 849 molecules) reveals that the hybrid representation (AUC 0.842) matches classical ECFP4 (AUC 0.868, $p = 0.111$, not significant) — positioning quantum-inspired methods as effective complementary tools alongside classical fingerprints. The quantum kernel applicability domain analysis identifies African NP compounds that fall outside the training distribution of existing activity prediction models. All representations, benchmark results, and analysis scripts are deposited on Zenodo and GitHub under open-source licences.

Author Contributions

Conceptualization and methodology, M.V.S.T. and S.G.N.E.; software, M.V.S.T. and J.-P.T.N., P.S.; validation, M.V.S.T., J.-P.T.N. and W.F.M.; formal analysis and investigation, M.V.S.T. and S.G.N.E.; resources, S.G.N.E. and W.F.M., P.S.; data curation, M.V.S.T.; writing–original draft preparation, M.V.S.T.; writing–review and editing, M.V.S.T., J.-P.T.N., P.S., W.F.M. and S.G.N.E.; visualization, M.V.S.T.; supervision, S.G.N.E. and W.F.M.; project administration, S.G.N.E. All authors read and agreed to the published version.

Data Availability

All data supporting the conclusions of this study are publicly available on Zenodo (DOI: https://doi.org/10.5281/zenodo.19608875, shared with the screening library dataset from Temgoua et al. (2026)) and mirrored at https://github.com/NanaEngo/Malaria_codesV2 under the MIT licence. The full deposit includes: (1) TFP feature matrices for all 19 849 library molecules (H_0/H_1 persistence diagrams and summary statistics); (2) TNE embeddings (192-element Tucker core descriptors, bond dimension 8, reconstruction errors); (3) quantum kernel matrices (8-qubit IQPEmbedding, PennyLane 0.45.1); (4) all benchmark CSV files (5-fold CV splits, AUC scores per descriptor); (5) the cross-paper H_1/RRS analysis dataset (`p3_polypharm_tfp_rrs.csv`); and (6) all analysis scripts.

Competing Interests

The authors declare no competing interests.

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References

D. Rogers and M. Hahn. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 2010. doi: 10.1021/ci100050t.

Maria Sorokina et al. COCONUT online: Collection of open natural products database. *Journal of Cheminformatics*, 13:2, 2021. doi: 10.1186/s13321-020-00478-9.

Jude Y. Betow, Ammar Qaseem, Boris D. Bekono, Yue Feng, Aurélien F. A. Moumbock, Conrad V. Simoben, Smith B. Babiaka, Solange A. Tanyi, Vanessa A. Shu, Ariane T. Ndi, Pascal Amoa Onguéné, Clovis S. Metuge, Simeon Akame, Donatus B. Eni, Cyril T. Namba-Nzanguim, Mathieu J. M. Tjegbe, Marianka N. L. Dikande, Patience T. Akeh, Sounders Balgah, Cyprian D. Meh, Akachukwu Ibezim, Idris F. Tabi, Yvette I. Malange, Bakoh Ndingkokhar, Leonel E. Njume, Kiran K. Telukunta, Said Amrani, Oyere T. Ebob, Wolfgang Sippl, Stefan Günther, and Fidele Ntie-Kang. African Natural Products Database (ANPDB): A resource for exploring the therapeutic potential of natural compounds from Africa. *Nucleic Acids Research*, 54(D1):D1336–D1344, January 2025. doi: 10.1093/nar/gkaf1186.

Godfrey Mayoka, Peter Mubanga Cheuka, Phanankosi Moyo, Godwin Akpeko Dziwornu, and Denzil Beukes. Value addition to African natural product-based drug discovery: barriers and opportunities. *Journal of Natural Products*, 88(8):2018–2028, July 2025. doi: 10.1021/acs.jnatprod.5c00446.

JunJie Wee and Jian Jiang. A review of topological data analysis and topological deep learning in molecular sciences. *arXiv Preprint*, September 2025. URL https://arxiv.org/abs/2509.16877.

Yuxi (Jaden) Long and Bruce R. Donald. Interpretable binding affinity prediction with persistent homology. *PLOS Computational Biology*, 21(6):e1013216, 2025. doi: 10.1371/journal.pcbi.1013216.

Wenbo Zhang et al. Task-specific pre-training for molecular property prediction. *Briefings in Bioinformatics*, 27(1):bbag010, January 2026. doi: 10.1093/bib/bbag010.

Guillaume Simeon and Gianni de Fabritiis. TensorNet: Cartesian tensor representations for efficient learning of molecular potentials. In *Advances in Neural Information Processing Systems*, 2023. URL https://openreview.net/pdf?id=BEHlPdBZ2e.

Richard M. Milbradt, Shuo Sun, Christian B. Mendl, Johnnie Gray, and Garnet K.-L. Chan. Efficient application of tensor network operators to tensor network states. *arXiv Preprint*, January 2026. URL https://arxiv.org/abs/2601.19650.

Yousra Farhani. Quantum machine learning: Where are we and what is next? *ACM Computing Surveys*, January 2026. doi: 10.1145/3764923.3777403.

Grier M. Jones, Viki Kumar Prasad, Ulrich Fekl, and Hans-Arno Jacobsen. Parametrized quantum circuit learning for quantum chemical applications. *Journal of Chemical Information and Modeling*, March 2026. doi: 10.1021/acs.jcim.6c00376.

Lohitaksh Badarala. Q-CaDD: accelerating in silico methodologies with quantum computation and machine learning for Epidermal growth factor receptor. *Scientific Reports*, 16:54321, March 2026. doi: 10.1038/s41598-026-44978-4.

Myke Vital Sao Temgoua, Jean-Pierre Tchapet Njafa, Penabei Samafou, and Wilfred Fon Mbacham. Ai-driven discovery of polypharmacological antimalarials from African natural product chemical space. *Journal of Chemical Information and Modeling*, 31(7):xxxx–xxxx, 2026. doi: pending. Paper 1: 65,856 molecules, dual-filter consensus docking.

John Preskill. Noisy intermediate-scale quantum: the long and the short of it. *Nature Reviews Physics*, 6:728–740, 2024. doi: 10.1038/s42254-024-00742-9.

K. Jacob, J. Clement, M. Arockiaraj, and A. Jindal. ChemGraphX: an open-source web tool for computing topological indices and entropy measures. *Journal of Cheminformatics*, 18:xx, February 2026. doi: 10.1186/s13321-025-01141-x.

Stalin Arulsamy, Kiruthiga Natarajan, Mohankumar Ramar, and Parasuraman Pavadai. An interactive visualization platform for explainable cheminformatics: Universal Molecular Fingerprint Highlighter (UMFH). *In Silico Research in Biomedicine*, March 2026.

Andac Demir, Baris Coskunuzer, Ignacio Segovia-Dominguez, Yuzhou Chen, Yulia Gel, and Bulent Kiziltan. ToDD: Topological compound fingerprinting in computer-aided drug discovery. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 27978–27993, 2022. URL https://arxiv.org/abs/2211.03808.

Hui Cang and Guo-Wei Wei. Topologynet: Topology based deep convolutional and multi-task neural networks for biomolecular property predictions. *Journal of Chemical Information and Modeling*, 58(2):189–197, 2018. doi: 10.1021/acs.jcim.7b00661.

Soham Mukherjee, Shreyas N. Samaga, Cheng Xin, Steve Oudot, and Tamal K. Dey. D-gril: End-to-end topological learning with 2-parameter persistence. In *42nd International Symposium on Computational Geometry (SoCG)*, volume 367, pages 79:1–79:15, 2026. doi: 10.4230/LIPIcs.SoCG.2026.79.

Alejandro Giraldo, Daniel Ruiz, Mariano Caruso, Javier Mancilla, and Guido Bellomo. Q2sar: A quantum multiple kernel learning approach for drug discovery. *arXiv preprint*, 2025. URL https://arxiv.org/abs/2506.14920.

Rahul Khorana. Topological feature compression for molecular graph neural networks. *arXiv preprint*, 2025. URL https://arxiv.org/abs/2508.07807.

David S. Wishart et al. DrugBank 5.0: a major update to the DrugBank database for 2018. *Nucleic Acids Research*, 46:D1074–D1082, 2018. doi: 10.1093/nar/gkx1037.

RDKit Contributors. RDKit: Open-source cheminformatics. https://www.rdkit.org/, 2024.

U. Bauer. Ripser: Efficient computation of vietoris–Rips persistence barcodes. *arXiv Preprint*, 2019. URL https://arxiv.org/abs/1908.02518.

GUDHI Project. GUDHI: Geometry and Understanding in Data Intelligence. https://gudhi.inria.fr/, 2024.

J. Kossaifi, Y. Panagakis, A. Anandkumar, and M. Pantic. TensorLy: Tensor learning in Python. *Journal of Machine Learning Research*, 20(26):1–6, 2019.

Xanadu Quantum Technologies. PennyLane: Quantum machine learning library. https://pennylane.ai/, 2024.

Jan H Jensen. Augmenting genetic algorithms with deep neural networks for exploring the chemical space. *Chemical Science*, 10(12):3567–3572, 2019. doi: 10.1039/c8sc05372c.

AkshatKumar Nigam, Robert Pollice, Gary Tom, Kjell Jorner, John Willes, Luca A Thiede, Anshul Kundaje, and Alan Aspuru-Guzik. Tartarus: A benchmarking platform for realistic and practical inverse molecular design. In *Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS 2023)*, 2023. URL https://arxiv.org/abs/2209.12487.